JIMMY WU

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EDUCATION

Washington University in St. Louis, Olin Business School
Ph.D. in Finance 2019 – 2026 (expected)
M.S. in Finance 2016 – 2017

University of California, Berkeley
B.A. in Economics and Statistics (Double Major, High Honors in Economics) 2011 – 2014

RESEARCH INTERESTS

Empirical Asset Pricing, Behavioral Finance, Large Language Models, Artificial Intelligence and Machine Learning

WORKING PAPERS

News Perplexity *Job Market Paper*

Existing research suggests that investors inefficiently process text that induces cognitive strain. This paper challenges that view. Drawing on dual-process theory, I hypothesize that certain strain-inducing text disrupts heuristic (System 1) intuition and triggers deliberate (System 2) analysis. To capture such text, I introduce news perplexity—a metric derived from large language models—and find that high perplexity news significantly mitigates market underreaction. Validating the mechanism, high perplexity news coincides with a spike in EDGAR search traffic, which is consistent with the notion that investors respond to the cognitive trigger by acquiring more information. These findings provide the first evidence that beneficial cognitive strain can enhance market efficiency by forcing deeper information processing.

Chronologically Consistent Large Language Models *with Songrun He, Linying Lv, Asaf Manela*
Revise and Resubmit: Journal of Financial Economics

Large language models are increasingly used in social sciences, but their training data can introduce lookahead bias and training leakage. A good chronologically consistent language model requires efficient use of training data to maintain accuracy despite time-restricted data. Here, we overcome this challenge by training a suite of chronologically consistent large language models, ChronoBERT and ChronoGPT, which incorporate only the text data that would have been available at each point in time. Despite this strict temporal constraint, our models achieve strong performance on natural language processing benchmarks, outperforming or matching widely used models (e.g., BERT), and remain competitive with larger open-weight models. Lookahead bias is model and application-specific because even if a chronologically consistent language model has poorer language comprehension, a regression or prediction model applied on top of the language model can compensate. In an asset pricing application predicting next-day stock returns from financial news, we find that ChronoBERT and ChronoGPT's real-time outputs achieve Sharpe ratios comparable to a much larger Llama model, indicating that lookahead bias in this setting is modest. Our results demonstrate a scalable, practical framework to mitigate training leakage, ensuring more credible backtests and predictions across finance and other social science domains.

The True Colors of Money: Racial Representation and Asset Management *with Lina Han, Xing Huang, Ohad Kadan*
Conditionally Accepted: Review of Financial Studies

We examine the role of race and ethnicity in the mutual fund context at two distinct levels. At the fund manager level, we document a co-racial tilt – funds managed by minority-dominant (White-dominant) teams allocate larger portfolio weights to minority-led (White-led) firms. This tilt is not associated with superior performance. It diminishes as fund managers gain experience, suggesting the presence of inaccurate statistical discrimination. At the investor level, we find that minority-led funds are penalized similarly to White-dominant funds for poor performance, but are not rewarded as much for superior performance. Overall, our results uncover race-related investment choices at both levels.

Instruction Tuning Chronologically Consistent Language Models *with Songrun He, Linying Lv, Asaf Manela*

We introduce a family of chronologically consistent, instruction-following large language models to eliminate lookahead bias. Each model is trained only on data available before a clearly defined knowledge-cutoff date, ensuring strict temporal separation from any post-cutoff data. The resulting framework offers (i) a simple, conversational chat interface, (ii) fully open, fixed model weights that guarantee replicability, and (iii) a conservative lower bound on forecast accuracy, isolating the share of predictability that survives once training leakage is removed. Together, these features provide researchers with an easy-to-use generative AI tool useful for a wide range of prediction tasks that is free of lookahead bias.

TEACHING & RESEARCH EXPERIENCE

Guest Lecturer

Behavioral Finance	Fall 2023

Teaching Assistant

Behavioral Finance – Prof. Xing Huang	Fall 2023/2022/2021/2020, Spring 2024/2023/2022
Fintech: Methods and Practice – Prof. Andreas Neuhierl	Spring 2022/2021
Introduction to Blockchain and Cryptocurrencies – Prof. Ohad Kadan	Fall 2021
Seminar in Fintech – Prof. Ohad Kadan, Prof. Asaf Manela	Fall 2020
Options and Futures – Prof. Thomas Maurer	Fall 2018/2016, Summer 2018
Derivative Securities – Prof. Thomas Maurer	Summer 2018
Investment Theory – Prof. Thomas Maurer	Summer 2018

Research Assistant

RA for Prof. Asaf Manela	2019 – 2020
Pre-doc RA for Prof. Ohad Kadan, Prof. Asaf Manela, Prof. Xing Huang	2018 – 2019
Master's RA for Prof. Thomas Maurer	2016 – 2017

INDUSTRY EXPERIENCE

J.P. Morgan, St. Louis, MO – *Analyst, Asset & Wealth Management*	January 2018 – August 2018
Buckingham Asset Management, St. Louis, MO – *Investment Strategist*	June 2017 – December 2017
Smith Moore, St. Louis, MO – *Project Intern*	April 2017 – June 2017
Smith Moore, St. Louis, MO – *Student Practicum Consultant*	August 2016 – December 2016
Troutman Sanders, Hong Kong – *Winter Law Associate*	January 2016
Wong & Associates, Oakland, CA – *Paralegal Assistant*	October 2014 – February 2015
Securities and Futures Commission, Hong Kong – *Summer Intern*	June 2012 – August 2012

REFERENCES

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PERSONAL INFORMATION

Nationality: U.S. Citizen, Chinese (Hong Kong)
Languages: Fluent in English, Mandarin Chinese and Cantonese
Computing: Python, R, Stata, Julia, Matlab, HPC
Previous Certifications: FINRA Series 7, Series 63, CFA Level I